Circular Saw

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: ELECTRIC CIRCULAR SAW

What it is used for

An electric circular saw uses a rotating toothed blade to cut suitable materials along a straight or guided line. Correct blade selection, workpiece support, guard function, and controlled cutting technique help produce accurate cuts while reducing kickback, laceration, and flying-debris hazards.

Main parts

- Electric motor drives the circular blade.
- Circular blade cuts the workpiece.
- Fixed and retractable blade guards cover the blade where designed.
- Base plate or shoe supports the saw on the material.
- Depth adjustment sets blade exposure below the base plate.
- Bevel adjustment sets the cutting angle where fitted.
- Handle and trigger provide grip and motor control.
- Power cable or battery supplies electrical energy.
- Spindle lock supports blade changes where provided.
- Guide or fence helps maintain a straight cutting line where fitted.

Before you start

1. Only trained personnel should operate the electric circular saw.
2. Inspect the saw, blade, guards, base plate, trigger, cable or battery, and adjustment locks for damage.
3. Select a sharp, undamaged blade suitable for the material, cutting task, and saw speed rating.
4. Confirm that the blade is installed in the correct direction and that the arbor and flange are compatible.
5. Check that the lower guard moves freely and returns to its protective position without sticking.
6. Set cutting depth and bevel angle with the saw disconnected from power or the battery removed.
7. Secure the workpiece on stable supports, keeping the intended cut path clear of clamps and the supporting surface.
8. Wear safety glasses or a face shield, hearing protection, and suitable respiratory protection where dust is generated; secure loose clothing and long hair.

How to use

1. Mark the cutting line and confirm that the workpiece is supported on both sides as appropriate without pinching the blade.
2. Check the planned cut for hidden nails, screws, wiring, or other obstructions.

3. Adjust the cutting depth so the blade extends only as far below the workpiece as recommended by the manufacturer.
4. Set the bevel angle and guide or fence if the cut requires them, then lock all adjustments.
5. Hold the saw firmly with both hands using the designated handles and position your body outside the likely kickback path.
6. Keep the blade clear of the workpiece and start the saw before bringing it into the cut.
7. Allow the blade to reach operating speed, then advance smoothly along the marked line without forcing the saw.
8. Keep the base plate flat against the workpiece and maintain control throughout the cut.
9. Release the trigger at the end of the cut and hold the saw still until the blade stops completely.
10. Wait for the blade to stop before lifting the saw, setting it down, or checking the cut.
11. Disconnect power or remove the battery before changing the blade, adjusting the saw, or clearing debris.
12. Inspect the cut and the saw, then clean and store the tool safely.

⚠ Safety

- Never disable or tie back the blade guard; do not operate if it fails to return properly.
- Keep hands away from the blade path and never reach beneath the workpiece while cutting.
- Do not hold small workpieces by hand or cut material that is unstable or inadequately supported.
- Avoid twisting the saw or forcing the blade, as this can cause binding and kickback.
- Do not use a damaged, cracked, incorrect, or blunt blade.
- Disconnect power or remove the battery before blade changes, adjustments, or maintenance, and wait for the blade to stop.

Care & storage

- Remove dust and debris from the saw after disconnecting power or removing the battery.
- Inspect the blade, guard mechanism, base plate, cable or battery contacts, and adjustment locks regularly.
- Replace damaged or blunt blades with compatible blades according to the manual.
- Keep ventilation openings clear and store the saw in a dry, secure location.
- Arrange authorised service if the guard sticks, the trigger malfunctions, or abnormal vibration develops.

Common mistakes to avoid

- Starting a cut with the blade touching the workpiece.
- Using a blade with the wrong type, direction, or speed rating.
- Cutting unsupported material or allowing the blade to bind in the kerf.
- Lifting or setting down the saw before the blade has stopped completely.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.